

Chapter 10

Arbitrage Pricing Theory and Multifactor Models of Risk and Return

INVESTMENTS

THIRTEENTH EDITION BODIE, KANE,
MARCUS

Overview

- *Arbitrage* is the exploitation of security mispricing in such a way that risk-free profits can be earned.
- Most basic principle of capital market theory is that well-functioning security markets rule out arbitrage opportunities.
- Generalization of the security market line of the CAPM to gain richer insight into the risk–return relationship.
- *Arbitrage pricing theory (APT)*.

Multifactor Models: A Preview

- Recall stock variability may be decomposed into the following sources:
 - Market (systematic) risk.
 - Largely due to macroeconomic events.
 - Firm-specific (idiosyncratic) effects.
- However, risk premiums may depend on correlations with extra-market risk factors.
 - For example, inflation, interest rates, volatility, and so on.

Factor Models of Security Returns ¹

- **Single-factor** model of excess returns

$$R_i = E(R_i) + \beta_i F + e_i$$

$E(R_i)$ = expected excess return on stock i

β_i = sensitivity of firm i

F = deviation of the common factor from its expected value

e_i = nonsystematic components of returns

Factor Models of Security Returns ²

- Extra market sources of risk may arise from several sources.
 - For example, uncertainty about interest rates or inflation.
- **Multifactor models** posit that returns respond to several systematic risk factors, as well as firm-specific influences.
 - Useful in risk management applications.

Factor Models of Security Returns ³

$$R_i = E(R_i) + \beta_{iGDP}GDP + \beta_{iIR}IR + e_i$$

R_i = excess return on security i .

B_{GDP} = sensitivity of returns to GDP

B_{IR} = sensitivity of returns to interest rates

e_i = nonsystematic components of returns

Arbitrage

- Arbitrage opportunity exists when an investor can earn riskless profits without making a net investment.
- No investment → investors create large positions to obtain large profits
- All investors will want an infinite position in the risk-free arbitrage portfolio
- In efficient markets, profitable arbitrage opportunities will quickly disappear

Law of One Price

- **Law of One Price.**
 - States that if two assets are equivalent in all economically relevant respects, then they should have the same market price
 - Enforced by arbitrageurs; If they observe a violation, they will engage in arbitrage *activity*.
 - For example, shares of a stock sell for different prices on two different exchanges.
 - This bids up (down) the price where it is low (high) until the arbitrage opportunity is eliminated.

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Arbitrage Pricing Theory

- **Arbitrage pricing theory (APT)** was developed by Stephen Ross.
- Predicts a SML linking expected returns to risk, but the path is takes to the SML is quite different.
- APT relies on three key propositions:
 1. Security returns can be described by a factor model.
 2. There are sufficient securities to diversify away idiosyncratic risk.
 3. Well-functioning security markets do not allow for the persistence of arbitrage opportunities.

APT and Well-Diversified Portfolios

$$R_P = E(R_P) + \beta_P F + e_P$$

where $F = \text{Systematic Risk}$

$$E(R_P) = \sum w_i E(R_i)$$

$$\beta_P = \sum w_i \beta_i$$

$$e_P = \sum w_i e_i$$

For a well-diversified portfolio,

- $e_P \rightarrow 0$ as the number of securities increases
- and their associated weights decrease

Well-Diversified Portfolio ₁

- From Chapter 8, $\sigma_P^2 = \beta_P^2 \sigma_F^2 + \sigma^2(e_P)$
- A well-diversified portfolio is defined as one that is diversified over a large number of securities so that for practical purposes the nonsystematic variance, $\sigma^2(e_P)$, is negligible.
- Because the expected value of e_P , $\sum w_i e_i$, for any well-diversified portfolio is zero, and its variance also is effectively zero. One can conclude that any realized value of e_P will be virtually zero.

Well-Diversified Portfolio ₂

- From Chapter 8, because e_i s are uncorrelated, variance of the equally weighted portfolio of firm-specific components is

$$\begin{aligned}\sigma^2(e_P) &= \text{Variance}\left(\sum w_i e_i\right) = \text{Var}\left(\sum_{i=1}^n \left(\frac{1}{n}\right) (e_i)\right) \\ &= \sum_{i=1}^n \left(\frac{1}{n}\right)^2 \sigma^2(e_i) = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{n} \sigma^2(e_i)\right) = \frac{1}{n} \bar{\sigma}^2(e)\end{aligned}$$

where $\bar{\sigma}^2(e)$ is the average of the firm-specific variances.

- Because this average is independent of n , when n gets large, it becomes negligible.

Figure 10.1 Scatter Diagram for Returns

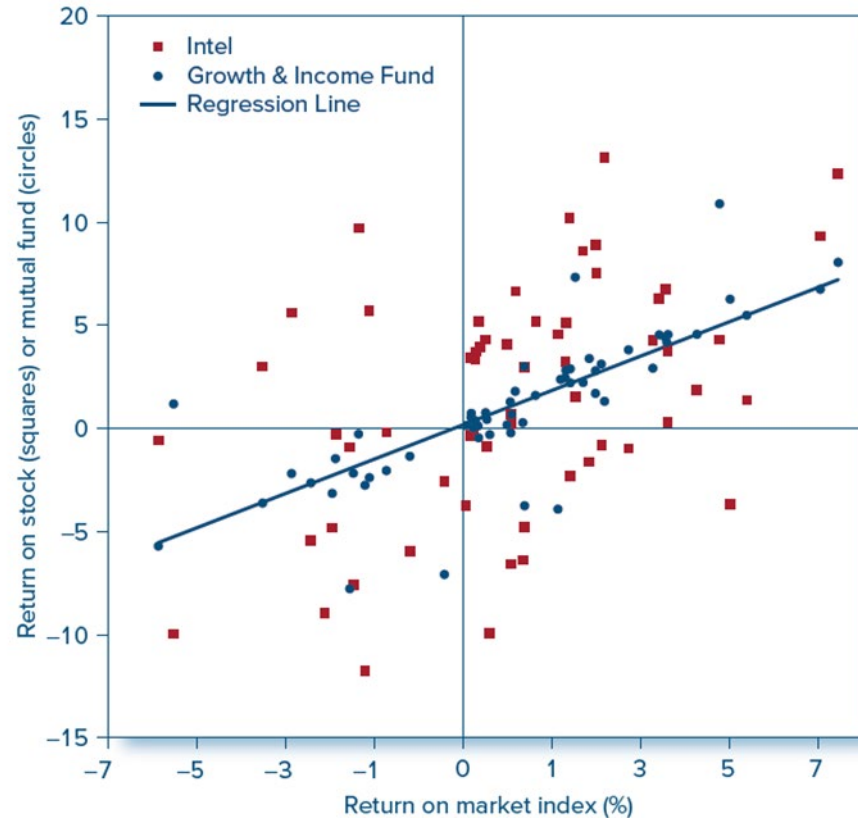
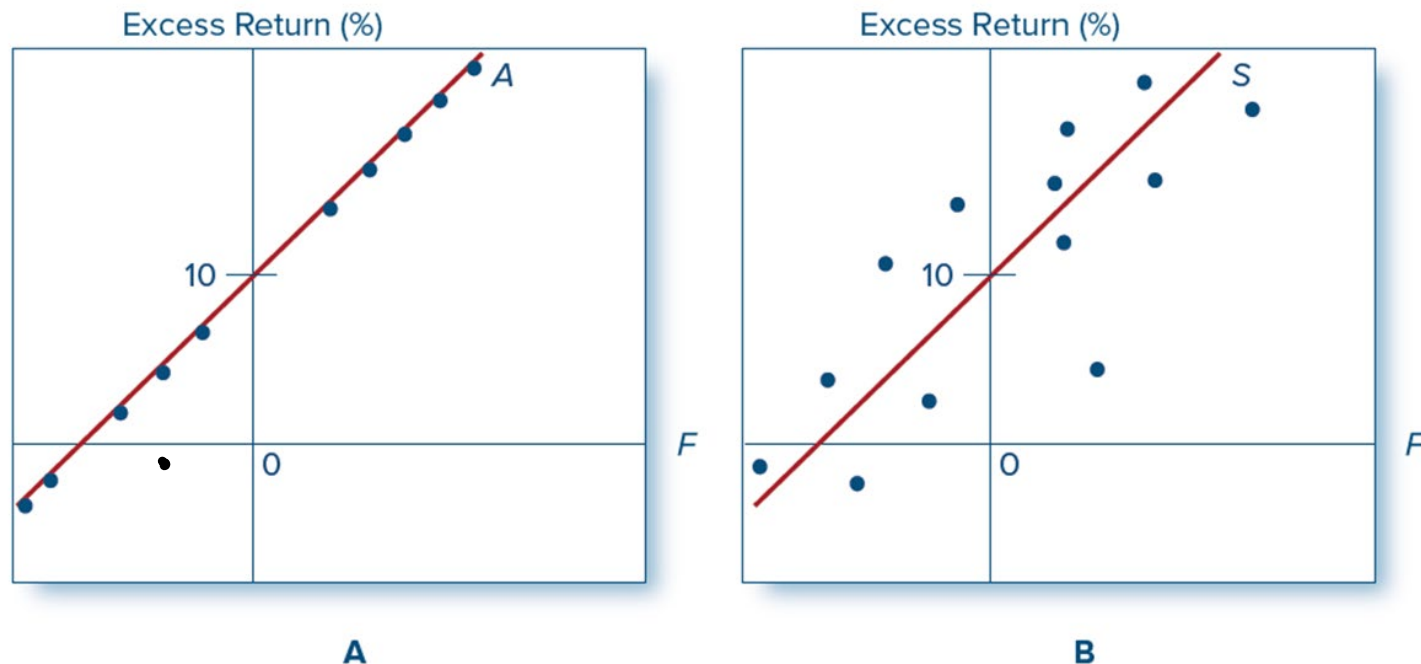


Figure 10.1 Scatter diagram for a single stock (Intel) and a diversified mutual fund (Vanguard Growth and Income). The fund exhibits much smaller scatter around the regression line.

Figure 10.2 Excess Returns as a Function of the Systematic Factor

$$r_P = E(r_P) + \beta_P F \quad \text{if } \beta_A = \beta_S = 1$$



Panel A, Well-diversified portfolio

Panel B, Single stock

Only factor risk should command a risk premium in market equilibrium

Figure 10.2 Excess returns as a function of the systematic factor: **Panel A**, Well-diversified portfolio A; **Panel B**, Single stock (S).

The SML of the APT

- All well-diversified portfolio with the same beta must have the same expected return.
- Generally, for any well-diversified P , the expected excess return must be:

$$E(R_p) = \beta_p E(R_M)$$

- Risk premium on portfolio P is the product of its beta and the risk premium of the market index.
- SML of the CAPM must also apply to well-diversified portfolios.

Figure 10.4 An Arbitrage Opportunity

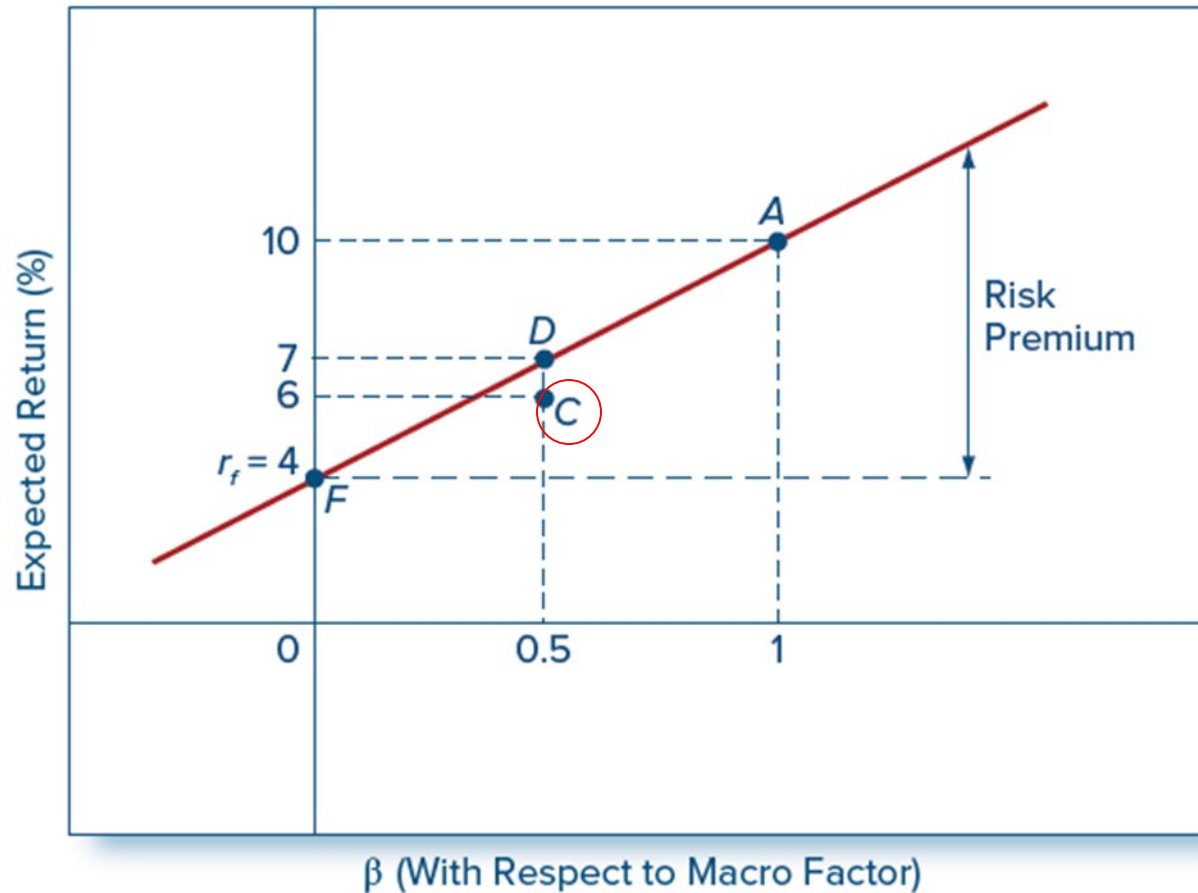


Figure 10.4 An Arbitrage Opportunity

- Construct a new portfolio D with half of portfolio A and half of the risk-free asset.
- D's beta = $.5 \times 0 + .5 \times 1.0 = .5$
- D's expected return is
$$(.5 \times 4 + .5 \times 10) = 7\% (>6\%)$$
- Sell short C and buy D can generate risk-free profit.
- Risk premiums must be proportional to beta

No-Arbitrage Equation of APT ¹

- To rule out arbitrage, the expected return on all well-diversified portfolio must lie on the straight line from the risk-free asset in Figure 10.4 (i.e., α must equal zero)
- Now consider the market index portfolio, M , as a well-diversified portfolio.
- The index portfolio, M , must be on the SML and the beta of the index portfolio is 1
- The intercept is r_f and the slope is $E(r_M) - r_f$, implying the equation of the line is

$$E(r_p) = r_f + \beta_p [E(r_M) - r_f]$$

No-Arbitrage Equation of APT ²

- We obtain the same expected return-beta relationship without requiring assumptions about either investor preferences or access to the all-inclusive (and elusive) market portfolio. (Note that R_M is the return of single factor)
- Accordingly, APT has more flexibility than does the CAPM since problems associated with an unobservable market portfolio are not a concern.

A Brief Summary: APT in a Single-factor Security Market

- In a single-factor security market, excess return,

$R_i = r_i - r_f$, can be represented as

$$R_i = \alpha_i + \beta_i R_M + e_i$$

- One can construct a well-diversified portfolio with zero firm-specific risk (the portfolio is still risky since the R_M is random)

$$R_i = \alpha_i + \beta_i R_M$$

- To rule out arbitrage, α s must equal zero.

$$R_p = \beta_p R_M$$

$$r_p - r_f = \beta_p (r_M - r_f)$$

- Taking expectation

$$E(r_p) = r_f + \beta_p [E(r_M) - r_f]$$

APT and CAPM

APT

- Built on the foundation of well-diversified portfolios.
- Cannot rule out a violation of the expected return–beta relationship for any particular asset.
- Does not assume investors are mean–variance optimizers.
- Uses an observable market index.

CAPM

- Model is based on an inherently unobservable “market” portfolio.
- Provides unequivocal statement on the expected return–beta relationship for all securities.

A Multifactor APT ¹

- To this point, we have examined the APT in a one-factor world.
- However, there are several sources of systematic risk and exposure to these factors will affect a stock's appropriate expected return.
- APT can be generalized to accommodate these multiple sources of risk.
- Assume a two-factor model:

$$R_i = E(R_i) + \beta_{i1}F_1 + \beta_{i2}F_2 + e_i$$

A Multifactor APT ₂

- Benchmark portfolios in the APT are factor portfolios.
- $\beta=1$ for one of the factors and $\beta=0$ for any other factors
- Factor portfolios track a particular source of macroeconomic risk, but are uncorrelated with other sources of risk
 - Referred to as a “tracking portfolio”

Fama-French (FF) Three-Factor Model

$$R_{it} = \alpha_i + \beta_{iM} R_{Mt} + \beta_{iSMB} SMB_t + \beta_{iHML} HML_t + e_{it}$$

SMB = Small minus big (the return of a portfolio of small stocks in excess of the return on a portfolio of large stocks)

HML = High minus low (the return of a portfolio of stocks with a high book-to-market ratio in excess of the return on a portfolio of stocks with a low book-to-market ratio)

Estimating and Implementing a Three-Factor SML

- Begin by estimating Amazon's beta on each of the FF factors
 - See Equation 10.10.
- Amazon's expected return in the multifactor model is lower due to the considerable hedging value it offers against the size and value risk factors.
 - This is ignored by the single-factor model.

Table 10.1 Estimates of Single-Index and Fama-French Regressions

	Single-index model		Three-factor model	
	Regression Coefficient	<i>t</i> -Statistic	Regression Coefficient	<i>t</i> -Statistic
Intercept (alpha, %)	−2.034	−0.991	−0.745	−0.363
$r_m - r_t$	2.030	4.785	1.771	4.074
SMB			0.369	0.487
HML			1.249	2.303
R-square	0.283		0.352	
Residual std dev (%)	15.171		14.675	

Table 10.1

Estimates of single-index and three-factor Fama-French regressions for U.S. Steel, monthly data, 5 years ending in 2021.

Source: Authors' calculations.

The Multifactor CAPM and the APT

- A multi-index CAPM inherits its risk factors from sources that a broad group of investors deem important enough to hedge
- The APT is largely silent on where to look for priced sources of risk

Smart Betas and Multifactor Models

Smart-beta *ETFs*

- Analogous to index ETFs, but instead provide exposure to specific characteristics.
- Fama-French five-factor model: Size (SMB), Value (HML), Operating Profitability (RMW), and Investment (CMA) in addition to Market.
- Momentum: Portfolio return that buys recent well-performing stocks and sells poorly performing ones.
- Other Factors: Volatility, as measured by the standard deviation of stock returns; Liquidity; and Dividend Yield.